



Intelligent

Document Verifier"



Proposed Solution:



Implement a system that leverages Optical Character Recognition (OCR) and Natural Language Processing (NLP) to extract and validate text from shipment-related documents (e.g., invoices, manifests). The solution aims to ensure that all necessary shipment details are present, even when information is embedded in scanned documents.



Workflow

- **Document Ingestion:**

Automatically ingest scanned PDFs or image files from a designated folder.

- **OCR Processing:**

Convert images into machine-readable text using OCR engines..

- **Document Ingestion:**

Apply NLP algorithms to parse the text and extract key fields (e.g., sender, destination, item description).

- **Document Ingestion:**

Compare extracted data against compliance criteria.



Techtack

Backend:

Python with libraries like Tesseract for OCR and spaCy for NLP.

Storage:

NoSQL databases like MongoDB for flexible document storage.

Frontend:

: Vue.js or Angular for interactive UI dashboards.

Cloud:

: AWS Lambda for serverless processing and S3 for document storage.



Viability and Feasibility:

Viability

- The combination of OCR and NLP is proven in document processing environments, making it a viable option for ensuring compliance.

Feasibility:

- While OCR accuracy can be affected by document quality, modern OCR engines and pre-processing techniques make the system feasible even with varied document inputs. Investment in training data and fine-tuning NLP models further enhances reliability.





Thank you

